

Radha Mastandrea

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Education

PhD in Physics

University of California, Berkeley

Berkeley, CA

Aug. 2021 - May 2025

Designated Emphasis in Computational and Data Science and Engineering

Thesis: Casting a Wide (Neural) Net: Using Machine Learning to Search for New Physics

Thesis Advisor: Ben Nachman

MPhil in Physics

University of Cambridge

Cambridge, UK

Oct. 2020 - Sep. 2021

Thesis: Investigating non-standard sources of parity violation at the LHC

Thesis Advisor: Christopher Lester

MASt in Physics

University of Cambridge

Cambridge, UK

Oct. 2019 - Jun. 2020

Distinguished Performance

Thesis: Search for new physics in $B_{(s)}^0 \rightarrow \mu^+ \mu^- \mu^+ \mu^-$ decays

Thesis Advisor: Valerie Gibson

B.S. in Physics

MIT

Cambridge, MA

Aug. 2015 - Jun. 2019

Thesis: Analyzing CMS Open Collider Data through Topic Modeling

Thesis Advisor: Jesse Thaler

Publications and Preprints

- [13] Chi Lung Cheng, Ranit Das, Runze Li, **Radha Mastandrea**, Vinicius Mikuni, Benjamin Nachman, David Shih, and Gup Singh. “Generator Based Inference (GBI)”. in: (May 2025). arXiv: [2506.00119](https://arxiv.org/abs/2506.00119) [hep-ph]
- [12] Rikab Gambhir, **Radha Mastandrea**, Benjamin Nachman, and Jesse Thaler. “Isolating Unisolated Upsilon with Anomaly Detection in CMS Open Data”. In: *Phys. Rev. Lett.* 135.2 (2025), p. 021902. DOI: [10.1103/vvv3-5kk1](https://doi.org/10.1103/PhysRevLett.135.021902). arXiv: [2502.14036](https://arxiv.org/abs/2502.14036) [hep-ph]
- [11] **Radha Mastandrea**, Benjamin Nachman, and Tilman Plehn. “Constraining the Higgs potential with neural simulation-based inference for di-Higgs production”. In: *Phys. Rev. D* 110 (5 Sept. 2024), p. 056004. DOI: [10.1103/PhysRevD.110.056004](https://doi.org/10.1103/PhysRevD.110.056004). URL: <https://link.aps.org/doi/10.1103/PhysRevD.110.056004>
- [10] Kehang Bai, **Radha Mastandrea**, and Benjamin Nachman. “Non-resonant anomaly detection with background extrapolation”. In: *Journal of High Energy Physics* 2024.4 (Apr. 2024). ISSN: 1029-8479. DOI: [10.1007/JHEP04\(2024\)059](https://doi.org/10.1007/JHEP04(2024)059). URL: [http://dx.doi.org/10.1007/JHEP04\(2024\)059](http://dx.doi.org/10.1007/JHEP04(2024)059)
- [9] Tobias Golling, Samuel Klein, **Radha Mastandrea**, Benjamin Nachman, and John Andrew Raine. “Morphing one dataset into another with maximum likelihood estimation”. In: *Phys. Rev. D* 108 (9 Nov. 2023), p. 096018. DOI: [10.1103/PhysRevD.108.096018](https://doi.org/10.1103/PhysRevD.108.096018). URL: <https://link.aps.org/doi/10.1103/PhysRevD.108.096018>
- [8] Tobias Golling, Gregor Kasieczka, Claudius Krause, **Radha Mastandrea**, Benjamin Nachman, John Andrew Raine, Debajyoti Sengupta, David Shih, and Manuel Sommerhalder. “The interplay of machine learning-based resonant anomaly detection methods”. In: *The European Physical Journal C* 84.3 (Mar. 2024). ISSN: 1434-6052. DOI: [10.1140/epjc/s10052-024-12607-x](https://doi.org/10.1140/epjc/s10052-024-12607-x). URL: <http://dx.doi.org/10.1140/epjc/s10052-024-12607-x>

- [7] Tobias Golling, Samuel Klein, **Radha Mastandrea**, and Benjamin Nachman. “Flow-enhanced transportation for anomaly detection”. In: *Phys. Rev. D* 107.9 (2023), p. 096025. DOI: [10.1103/PhysRevD.107.096025](https://doi.org/10.1103/PhysRevD.107.096025). arXiv: [2212.11285](https://arxiv.org/abs/2212.11285) [hep-ph]
- [6] **Radha Mastandrea** and Benjamin Nachman. “Efficiently Moving Instead of Reweighting Collider Events with Machine Learning”. In: *36th Conference on Neural Information Processing Systems: Workshop on Machine Learning and the Physical Sciences*. Dec. 2022. arXiv: [2212.06155](https://arxiv.org/abs/2212.06155) [hep-ph]
- [5] Gregor Kasieczka, **Radha Mastandrea**, Vinicius Mikuni, Benjamin Nachman, Mariel Pettee, and David Shih. “Anomaly detection under coordinate transformations”. In: *Phys. Rev. D* 107.1 (2023), p. 015009. DOI: [10.1103/PhysRevD.107.015009](https://doi.org/10.1103/PhysRevD.107.015009). arXiv: [2209.06225](https://arxiv.org/abs/2209.06225) [hep-ph]
- [4] Barry M. Dillon, **Radha Mastandrea**, and Benjamin Nachman. “Self-supervised anomaly detection for new physics”. In: *Phys. Rev. D* 106.5 (2022), p. 056005. DOI: [10.1103/PhysRevD.106.056005](https://doi.org/10.1103/PhysRevD.106.056005). arXiv: [2205.10380](https://arxiv.org/abs/2205.10380) [hep-ph]
- [3] Christopher G. Lester, **Radha Mastandrea**, Daniel Noel, and Rupert Tombs. “Hunting for vampires and other unlikely forms of parity violation at the Large Hadron Collider”. In: *JHEP* 08 (2022), p. 231. DOI: [10.1007/JHEP08\(2022\)231](https://doi.org/10.1007/JHEP08(2022)231). arXiv: [2205.09876](https://arxiv.org/abs/2205.09876) [hep-ph]
- [2] Brian T Cook, Deborah F Woods, Jessica D Ruprecht, Jacob Varey, **Radha Mastandrea**, Kaylee de Soto, Jacob F Harburg, Umaa Rebbapragada, and Ashish A Mahabal. “Tracing Milky Way substructure with an RR Lyrae hierarchical clustering forest”. In: *Monthly Notices of the Royal Astronomical Society* (Apr. 2022). stac1007. ISSN: 0035-8711. DOI: [10.1093/mnras/stac1007](https://doi.org/10.1093/mnras/stac1007). eprint: <https://academic.oup.com/mnras/advance-article-pdf/doi/10.1093/mnras/stac1007/43400845/stac1007.pdf>. URL: <https://doi.org/10.1093/mnras/stac1007>
- [1] Patrick T. Komiske, **Radha Mastandrea**, Eric M. Metodiev, Preksha Naik, and Jesse Thaler. “Exploring the Space of Jets with CMS Open Data”. In: *Phys. Rev. D* 101.3 (2020), p. 034009. DOI: [10.1103/PhysRevD.101.034009](https://doi.org/10.1103/PhysRevD.101.034009). arXiv: [1908.08542](https://arxiv.org/abs/1908.08542) [hep-ph]

Conference Talks

Isolating Unisolated Upsilons with Anomaly Detection in CMS Open Data

ML4Jets, Caltech

Aug. 2025

AD4HEP, Columbia University, Nevis Laboratories

Jun, 2025

Constraining the Higgs Potential Shape with Machine Learning

ML4Jets, LPNHE

Nov. 2024

DPF-PHENO, University of Pittsburgh

May 2024

A Survey of Machine Learning Methods for Anomaly Detection

Workshop on Machine Learning and High-Energy Physics, Campus Akademie, Vienna

Dec. 2023

The Interplay of Machine Learning-based Resonant Anomaly Detection Methods

Hammers & Nails, Ascona, Switzerland

Oct. 2023

ML4Jets, DESY

Nov. 2023

FETA: Flow-Enhanced Transportation for Anomaly Detection

APS April Meeting, Minneapolis

Apr. 2023

HEPSim2Real: Creating background templates with normalizing flows

ML4Jets, Rutgers University

Nov. 2022

Using symmetries to build better latent spaces for dijet representation learning

APS April Meeting, New York

Apr. 2022

Exploring the Parity of the Quark-Sector SME with Madgraph

Fourth Summer School on the Lorentz- and CPT-violating Standard-Model Extension, ICUSS

May 2021

Analyzing CMS Open Collider Data through Topic Modeling

BOOST Physics Workshop, *MIT*

Jul. 2019

Testing Parameterized Theories of General Relativity using Gravitational Waves

APS New England Section Meeting, *URI*

Dec. 2017

Seminars

Constraining the Higgs Potential Shape with Simulation-Based Inference

Rutgers NHETC Seminar

Dec. 2024

ATLAS ML Forum

Nov. 2024

CERN Theory ML Forum

Nov. 2024

CMS ML Forum

Nov. 2024

The Interplay of Machine Learning-based Anomaly Detection Methods

LHCb RTA-DPA Meeting

Nov. 2024

IAIFI Journal Club

Apr. 2024

RWTH HEP-ML Club

Jan. 2024

SLAC AI Seminar

Dec. 2023

FETA: Flow-Enhanced Transportation for Anomaly Detection

Fermilab AI Meetings

Jul. 2023

University College London HEP Seminars

Dec. 2022

Imperial College London HEP Seminars

Dec. 2022

UC Berkeley 4D Seminars

Oct. 2022

Honors and Awards

Apr. 2024

Berkeley Grad Slam Semi-Finalist

Aug. 2023 - May 2025

Templeton TEX Fellow

Apr. 2023

Citadel PhD Summit Attendee

Mar. 2023

APS GDS Impact Award

Mar. 2022 - Mar. 2023

APS DSECOP Fellow

Sep. 2021 - Aug. 2024

NSF Graduate Research Fellowship

Oct. 2019 - Jul. 2021

Marshall Scholarship

Jun. 2019

Joel Matthew Orloff Award for Outstanding Service

Jun. 2019

Phi Beta Kappa

Jun. 2019

Sigma Pi Sigma

Oct. 2019

FUTURE of Physics @ Caltech participant

May. 2018 - Sep. 2018

Heising-Simons @ MIT Physics Research Fellow

Summer Schools

Jan. 2025

GGI Theoretical Lectures on Fundamental Interactions, *GGI*

Jun. 2024

TASI, *UC Boulder*

Aug. 2023

SLAC Summer Institute, *SLAC*

Jul. 2023

CMS Open Data Workshop, *Fermilab*